

Hongchen (Steven) Yang

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Education

Brandeis University, Waltham, MA

Expected Dec 2026

B.S. in Computer Science GPA: 3.748 / 4.0

Dean's List (All Semesters); Provost's Summer Undergraduate Research Fellowship (Summer 2025)

Relevant Coursework: Computer Systems Design; Database Management; Operating Systems; Machine Learning; Theory of Computation

Research Experience

Smart & Scalable Data Systems (SSD) Lab, Researcher — Brandeis University

May 2025 – Present

- Study how workload shifts, skew, and burstiness affect RocksDB compaction and performance through LSM-tree literature review, source-code analysis, and benchmarking.
- Group tunable knobs by throughput, latency, CPU utilization, and amplification, prune dependent parameters, and investigate workload-aware selection using Bayesian optimization, tree-based surrogate models, and lightweight online learning.
- Build prototype benchmarks to evaluate tuning policies and inform *FluidLSM*, a self-adaptive LSM-based key-value store.

Data Science Intern (Independent Study) — Brandeis University

Jan 2026 – May 2026

- Design a controlled text-to-SQL benchmark with synthetic schemas, configurable fragmentation, Python ETL, provenance tracking, and join/query evaluation to measure the effects of heterogeneous and incomplete administrative data.

Publications

Dazhai Yang, Jiao Liu, **Hongchen Yang**, Hualin Liu, and Tianzhe Lou. “From Single-View to Multi-view: Learning Informative Graphs for Robust Subspace Segmentation.” *Advances in Artificial Intelligence, Electronic Instruments and Information Systems*, Springer, 2026, pp. 82–96. [DOI](#). *Third author; contributed in a supporting role.*

Teaching Experience

Teaching Assistant, Introduction to Database — Brandeis University

Jan 2026 – May 2026

- Lead office hours and review sessions for about 50 students and grade coursework covering SQL, ER modeling, normalization, indexing, and query optimization.

Internship Experience

Hefei City Cloud Data Center Co., Ltd. — Hefei, China

AI Development Intern

Jun 2026 – Aug 2026

- Built an end-to-end AI reimbursement workflow that ingests emails and attachments, recognizes documents, extracts structured fields, and prepares downstream data.
- Developed agent-memory and vector-retrieval nodes for memory lifecycle operations and scoped semantic search across multi-turn tasks.
- Added memory-resource APIs and UI, integrated terminology-retrieval and report-writing nodes, migrated workflows, and validated database, vector, and model-service integrations.

Shanghai Development Center of Computer Software Technology — Shanghai, China

Software Engineering Intern

May 2024 – Aug 2024

- Traced Supersonic AI's text-to-SQL workflow across translation, parsing, planning, and execution, and analyzed the RuoYi backend's Spring Boot, MyBatis, Redis, and Maven architecture.
- Used Linux tools to run services, inspect logs, and debug tasks; refactored Java service logic with streams and lambda expressions.

Projects

Blacklight Privacy Detection Tool — The Markup & Brandeis University

Aug 2025 – Dec 2025

- Contributed to development and validation of TikTok, X, and third-party tracking-pixel detection in Blacklight, a public privacy tool with more than 18 million scans.

Technical Skills

Languages: Java, C++, Python, Go, Rust, JavaScript, TypeScript, SQL

Frameworks & Systems: Spring Boot, MyBatis, RocksDB, PostgreSQL, MySQL, Redis, Docker, Linux, Git

Languages Spoken: Mandarin Chinese (Native), English (Fluent)